Practice workbook · Solving log and exponential equations with your graph

Using log laws, or switching between log and exponential form, to turn an equation into 'plotted curve = straight line', then solving it from the graph.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-GRA-09

1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

5 (a) Complete the table of values for $y = e^{3x-2}$ giving your answers to 2 decimal places.

x	0	0.25	0.5	0.75	1
y	0.14				2.72

(2)

(b) On the grid opposite, draw the graph of $y = e^{3x-2}$ for $0 \leqslant x \leqslant 1$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $3x = 2 + \ln(3 - x)$

(3)

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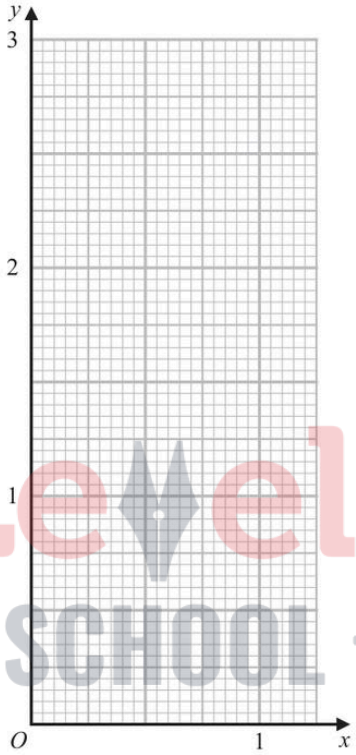
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1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 5 continued



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Turn over for a spare grid if you need to redraw your graph.

1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 5 continued

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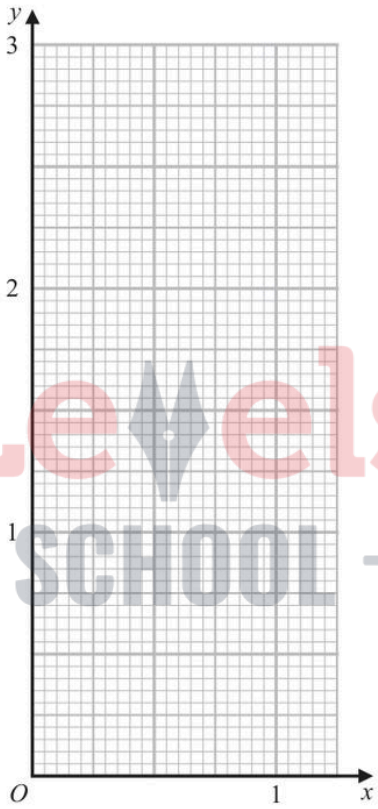
1. Question 5 · part c · 3 marks

2022 JUN · Paper 02 (regional R)

Worked-example retry · Attempt part c only.

Question 5 continued

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(Total for Question 5 is 7 marks)

1. Question 5 · part c · 3 marks · Working and answer

Working and answer

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2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

(4)

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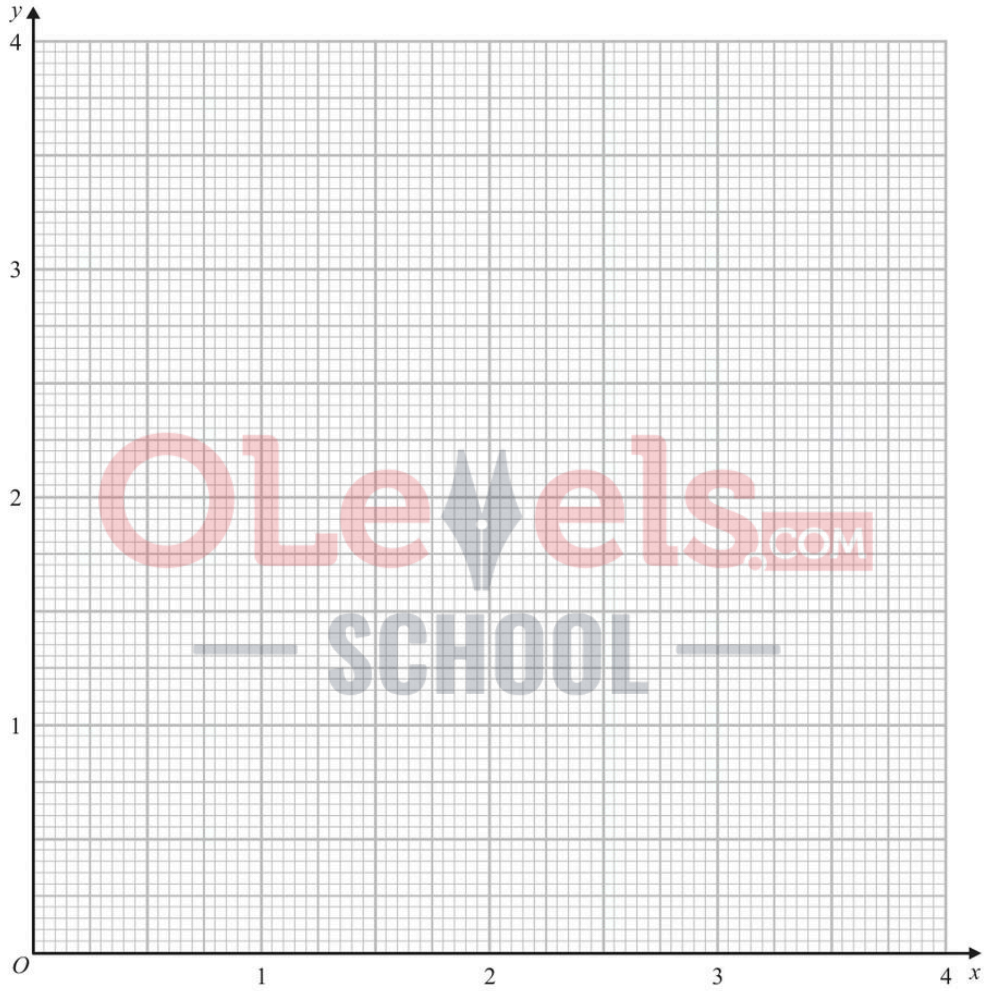
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2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 8 continued



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2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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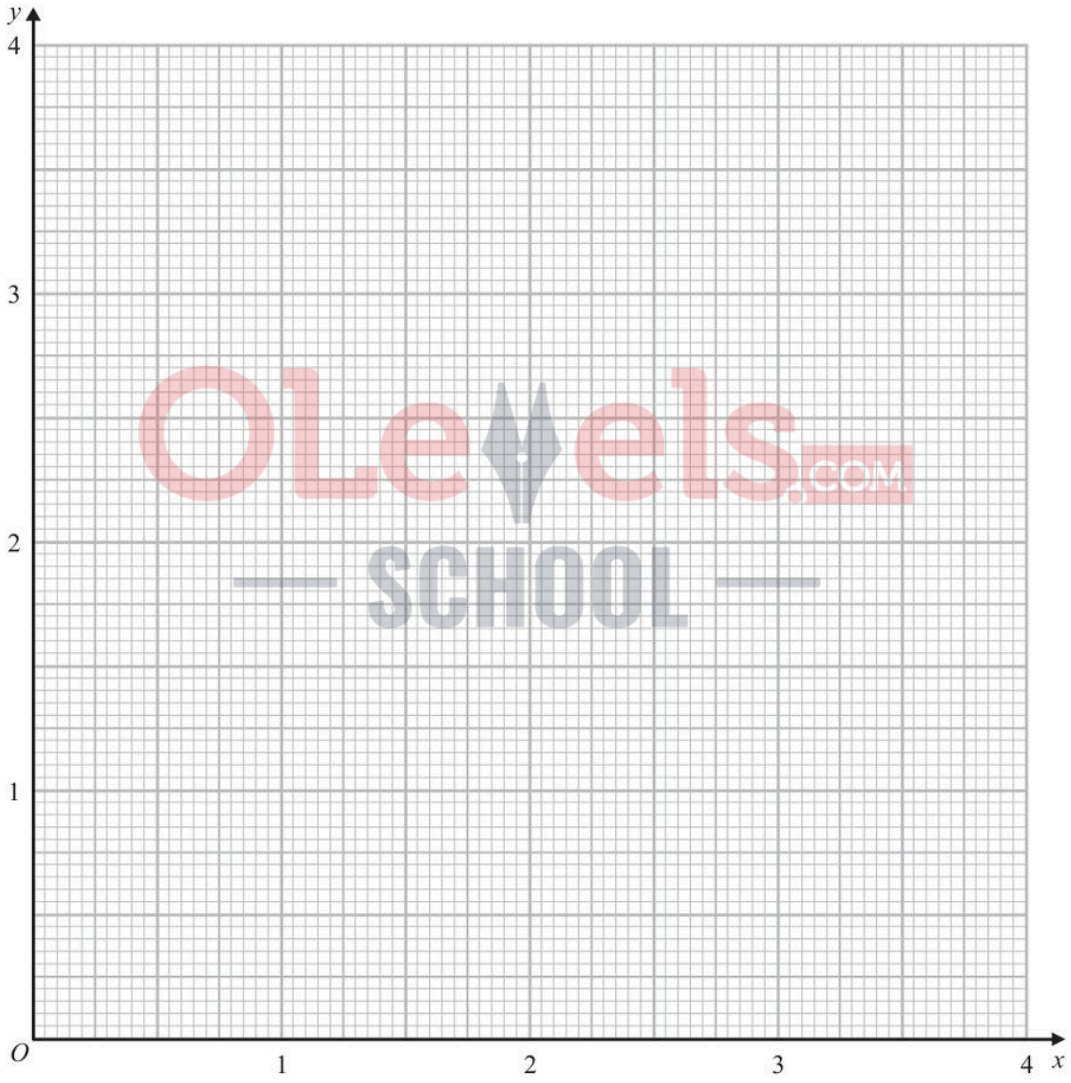
2. Question 8 · part d · 4 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part d only.

Question 8 continued

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(Total for Question 8 is 11 marks)

2. Question 8 · part d · 4 marks · Working and answer

Working and answer

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3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

7 (a) Complete the table of values for

$$y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$$

giving each value to 2 decimal places where appropriate.

x	–6	–5	–4	–3	–2	–1	0
y	4	3.59	3.26				2.5

(2)

(b) On the grid opposite, draw the graph of $y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ for $-6 \leqslant x \leqslant 0$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_2(2x+2)^3 + x + 3 = 0 \text{ in the interval } -6 \leqslant x \leqslant 0$$

(6)

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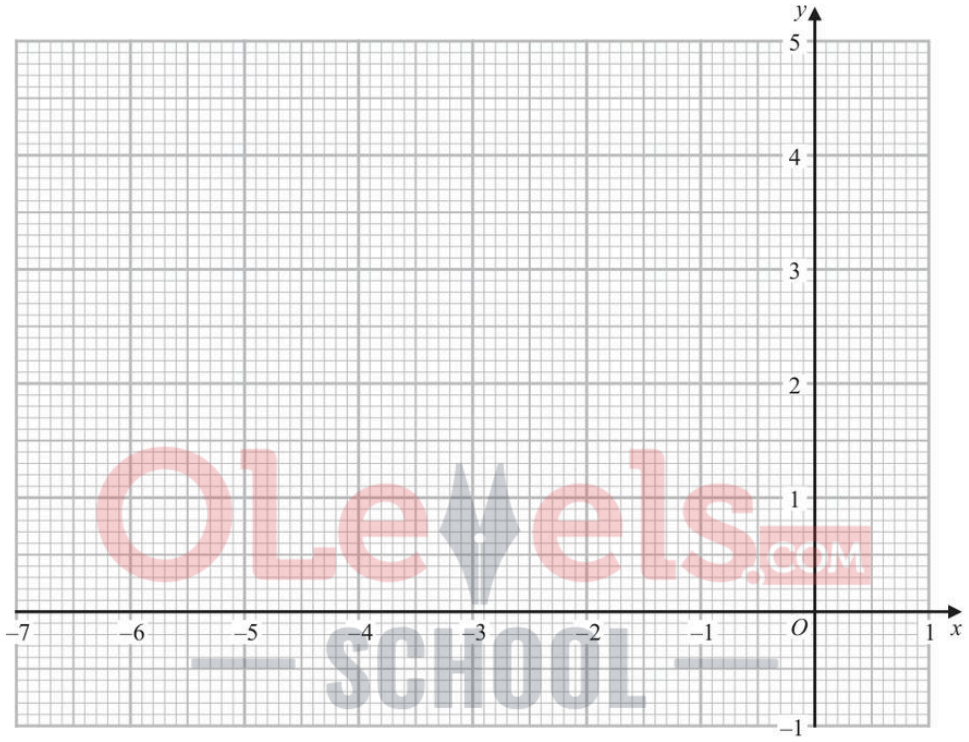
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3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued



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3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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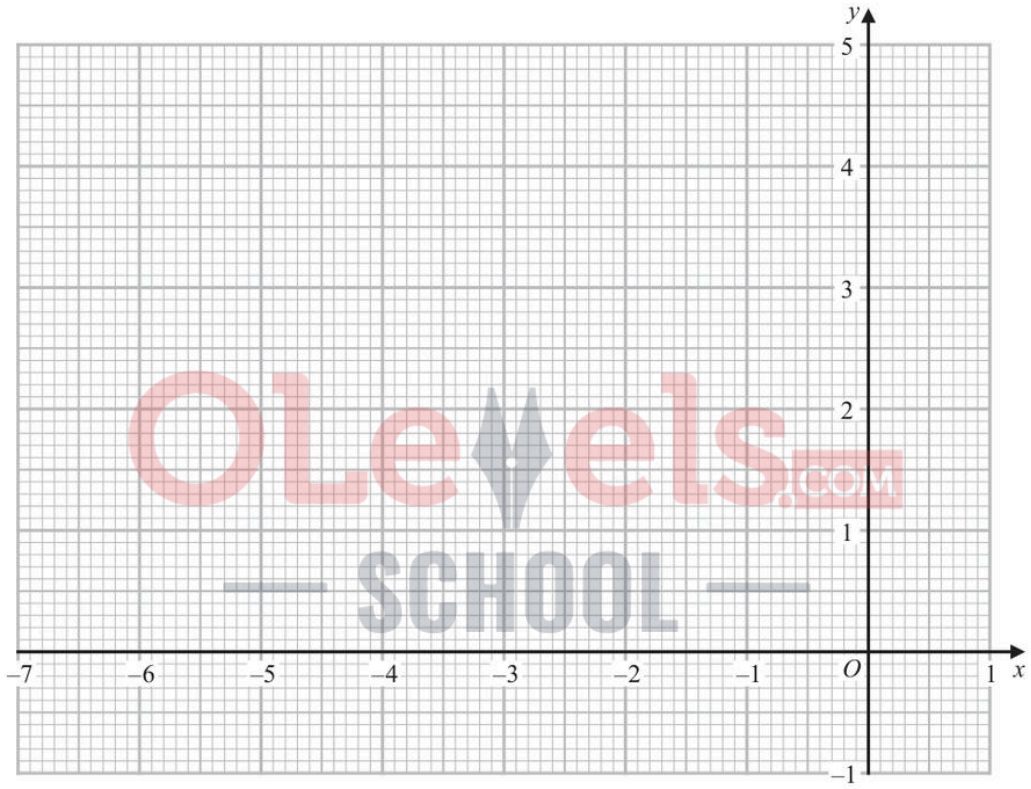
3. Question 7 · part c · 6 marks

2023 JAN · Paper 01

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 10 marks)

3. Question 7 · part c · 6 marks · Working and answer

Working and answer

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4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Complete the table of values for $y = \log_3(4-x)+3$ giving your answers to 2 decimal places.

x	0	0.5	1	1.5	2	2.5	3	3.5
y	4.26		4		3.63		3	2.37

(2)

(b) On the grid opposite, draw the graph of $y = \log_3(4-x)+3$ in the interval $0 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate,

to one decimal place, of the root of the equation $3^{2x-5} - (4-x)^3 = 0$ in the interval $0 \leqslant x \leqslant 3.5$

(6)

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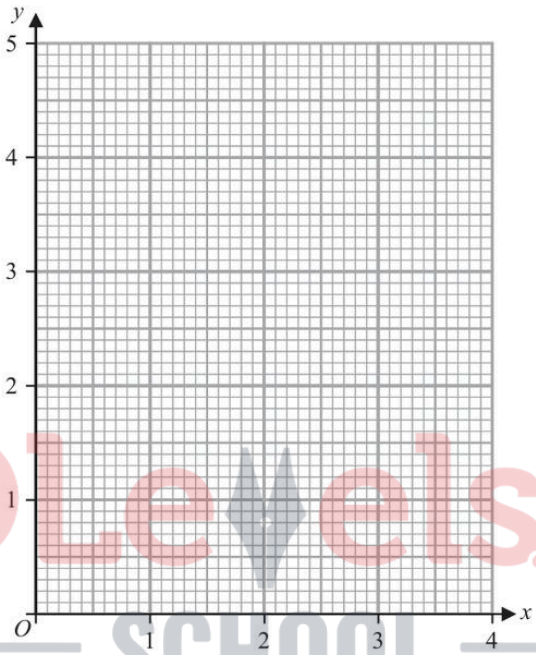
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4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued



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Turn over

4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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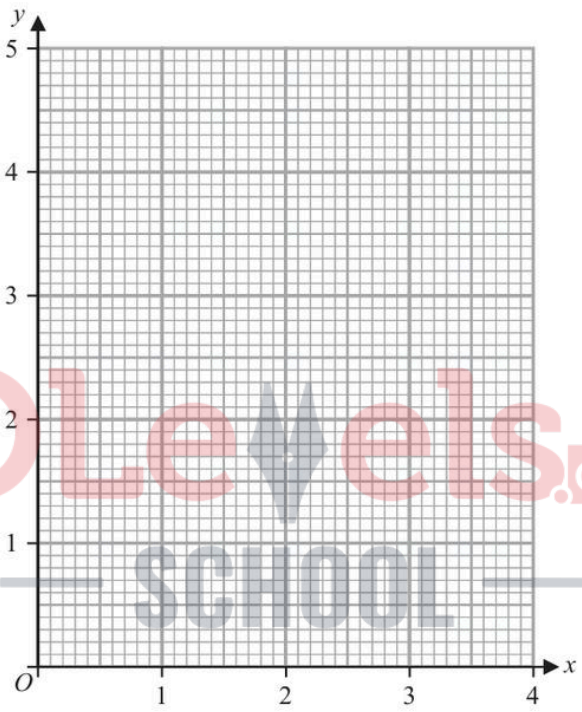
4. Question 7 · part c · 6 marks

2025 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 10 marks)

4. Question 7 · part c · 6 marks · Working and answer

Working and answer

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5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

4 (a) Complete the table of values for $y = \frac{x}{2} + 6e^{-2x} + 1$

giving your answers to one decimal place.

x	0	1	1.5	2	3	4	5	6
y	7		2.0			3.0		4.0

(2)

(b) On the grid opposite, draw the graph of $y = \frac{x}{2} + 6e^{-2x} + 1$ for $0 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on your graph, obtain estimates, to one decimal place, of the roots of the equation

$$2x + \ln(24 - 5x) = \ln 36$$

(5)

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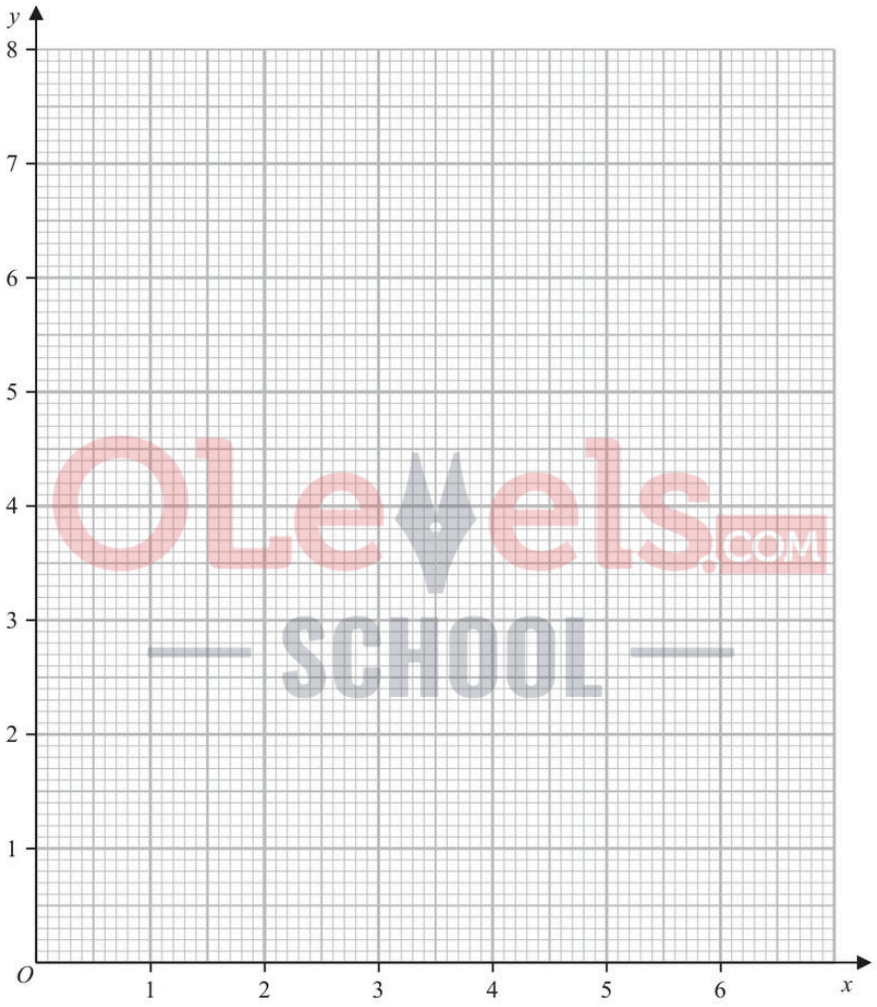
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5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued



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5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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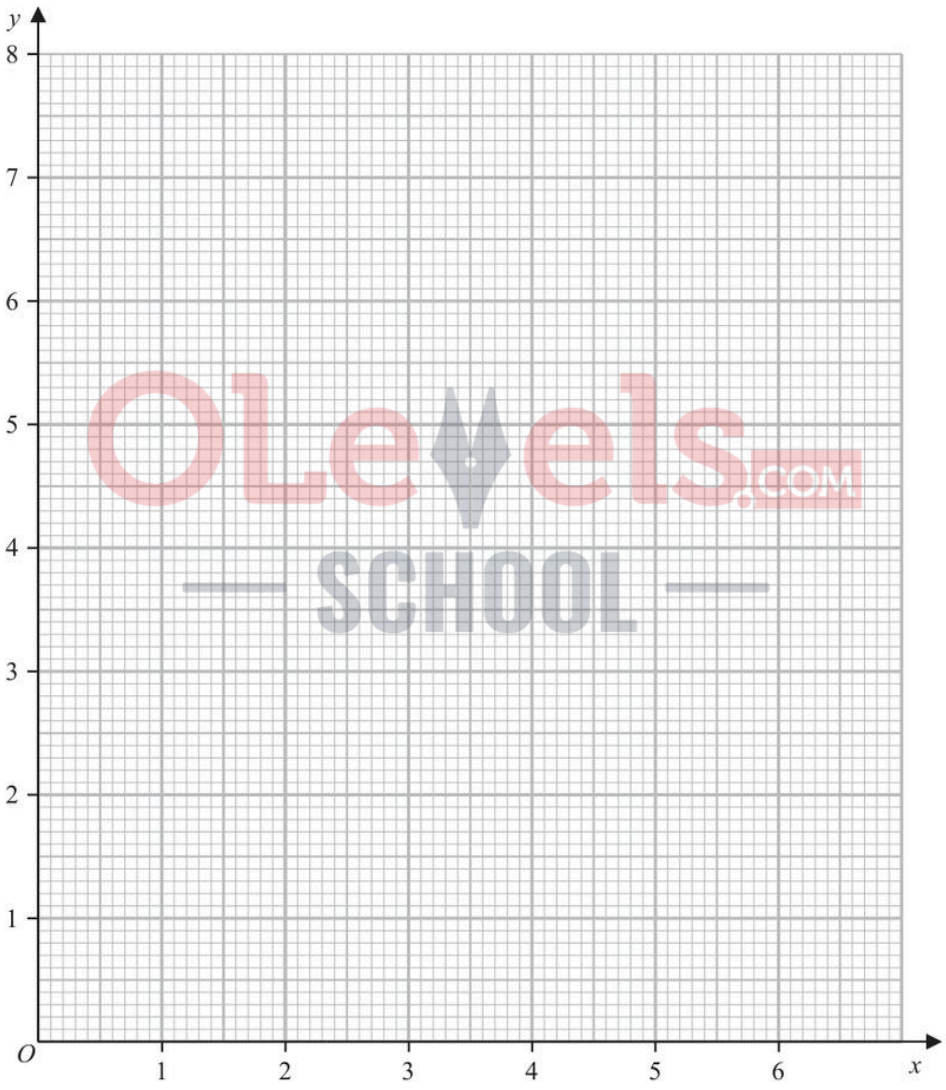
5. Question 4 · part c · 5 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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(Total for Question 4 is 9 marks)

5. Question 4 · part c · 5 marks · Working and answer

Working and answer

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6. Question 6 · part whole · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

6

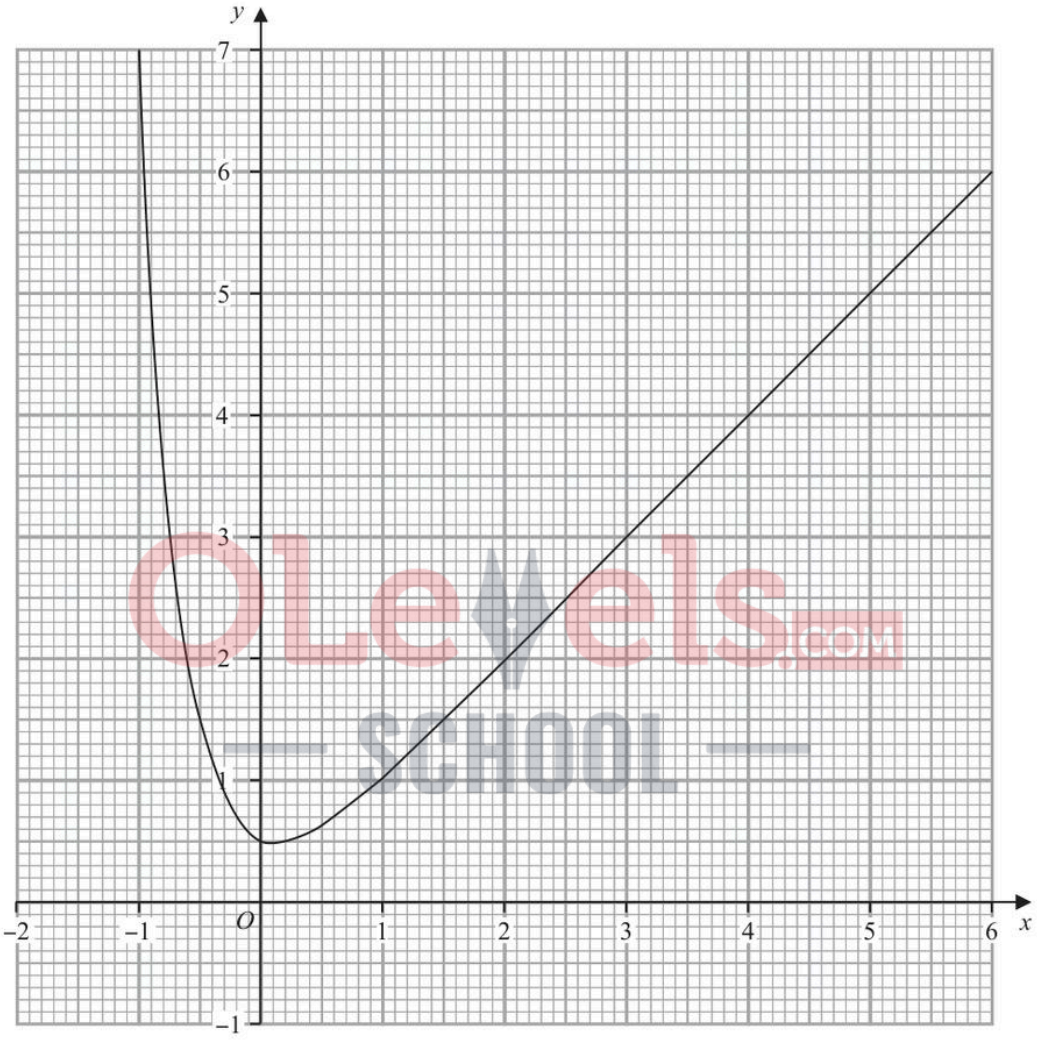


Figure 1

Figure 1 shows part of the graph of the curve with equation $y = x + 2^{-(4x+1)}$

By drawing a suitable straight line on the graph, obtain an estimate, to one decimal place, of the roots of the equation $\log_2(8 - 3x) + 4x = 0$ in the interval $-2 \leqslant x \leqslant 6$

(7)

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6. Question 6 · part whole · 7 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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Question 6 continued

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(Total for Question 6 is 7 marks)

6. Question 6 · part whole · 7 marks · Working and answer

Working and answer

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7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

(5)

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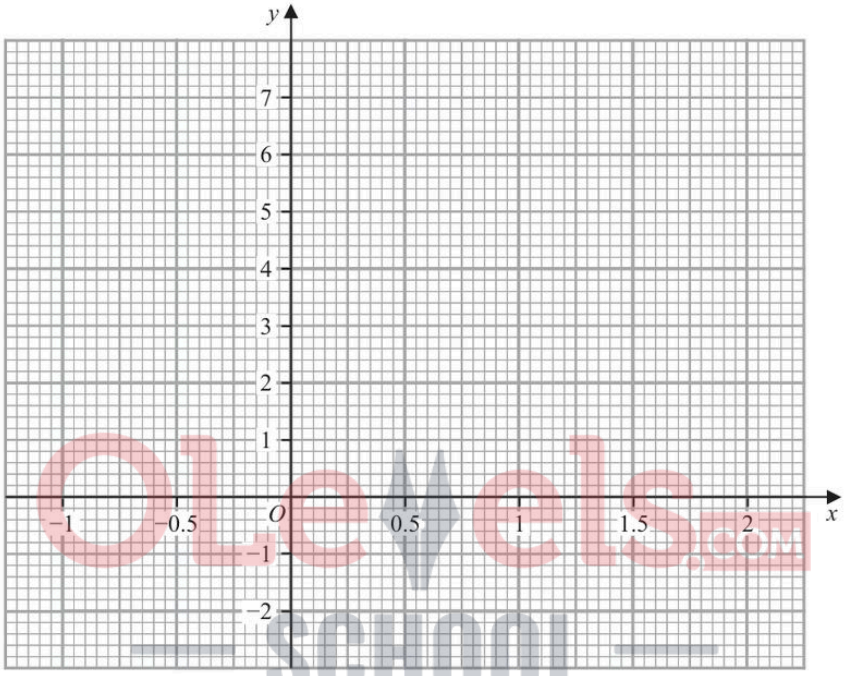
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7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

Question 8 continued



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7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

Question 8 continued

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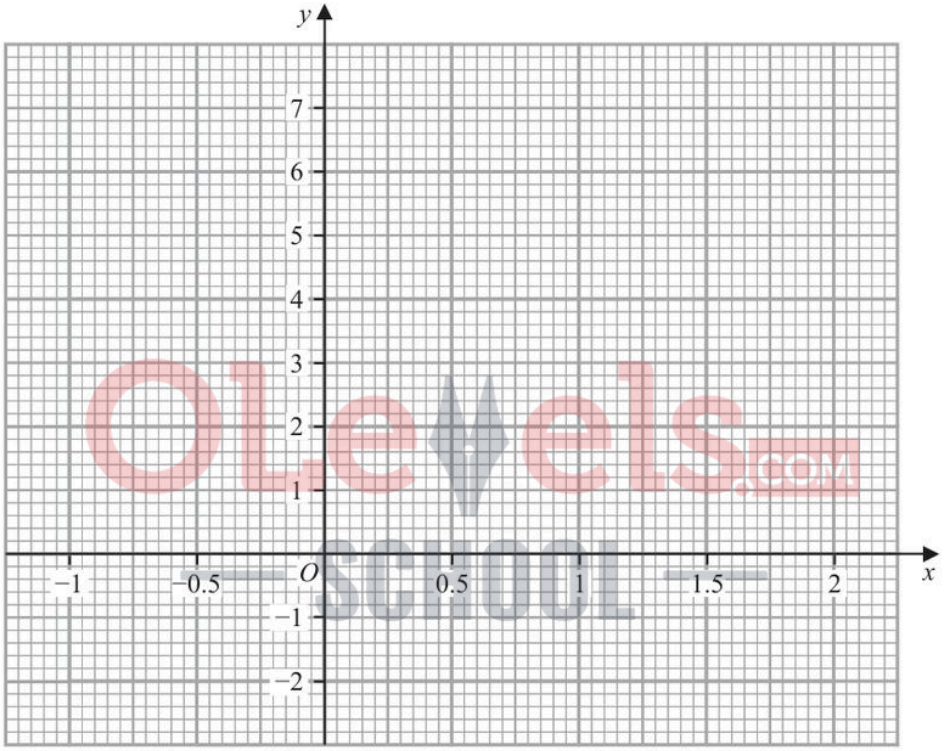
7. Question 8 · part d · 5 marks

2025 NOV · Paper 01

Further practice · Attempt part d only.

Question 8 continued

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(Total for Question 8 is 12 marks)

7. Question 8 · part d · 5 marks · Working and answer

Working and answer

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8. Question 8 · part c · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part c only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

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8. Question 8 · part c · 3 marks · Working and answer

Working and answer

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9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

7 (a) Complete the table of values for $y = 3^{\frac{x}{4}} + 2$

Give your answers to 2 decimal places where appropriate.

(2)

x	0	1	2	3	4	5
y	3	3.32				5.95

(b) On the grid opposite, draw the graph of

$$y = 3^{\frac{x}{4}} + 2 \qquad \text{for } 0 \leqslant x \leqslant 5$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_3(6 - 2x)^4 - x = 0$$

in the interval $0 \leqslant x \leqslant 5$

(5)

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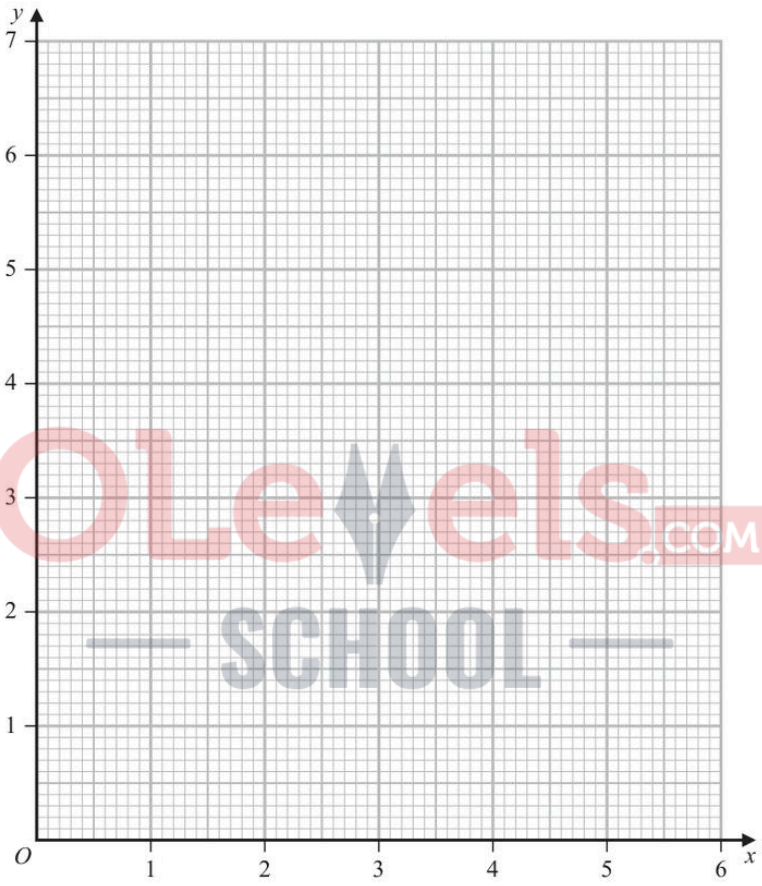
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9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued



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9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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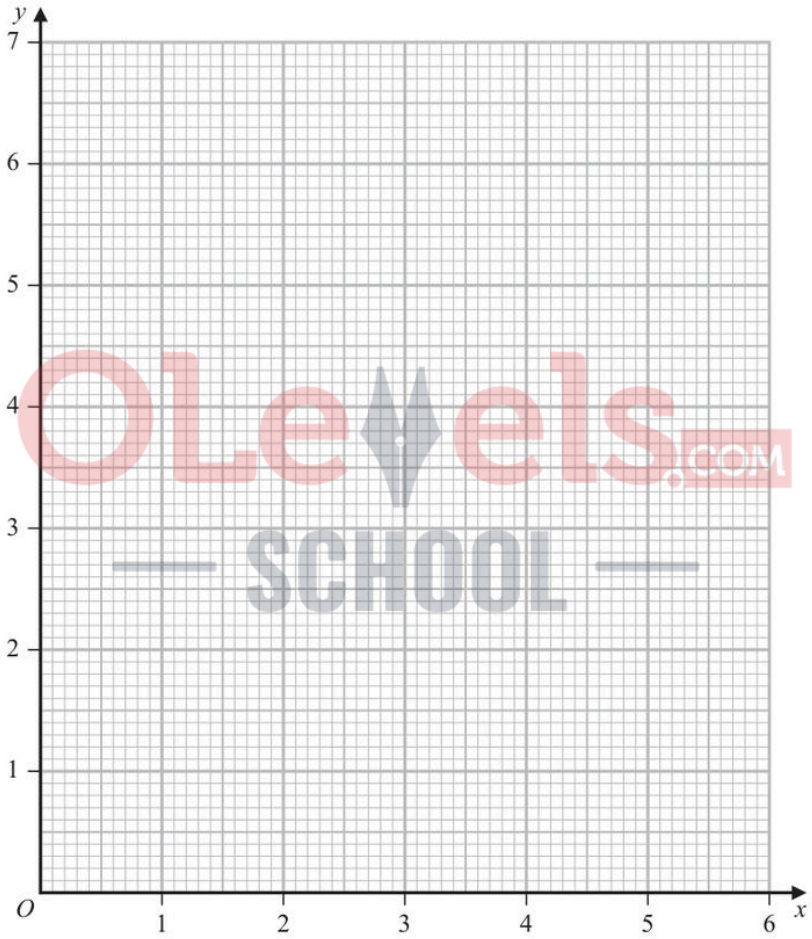
9. Question 7 · part c · 5 marks

2021 JUN · Paper 02

Further practice · Attempt part c only.

Question 7 continued

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(Total for Question 7 is 9 marks)

9. Question 7 · part c · 5 marks · Working and answer

Working and answer

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10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(4)

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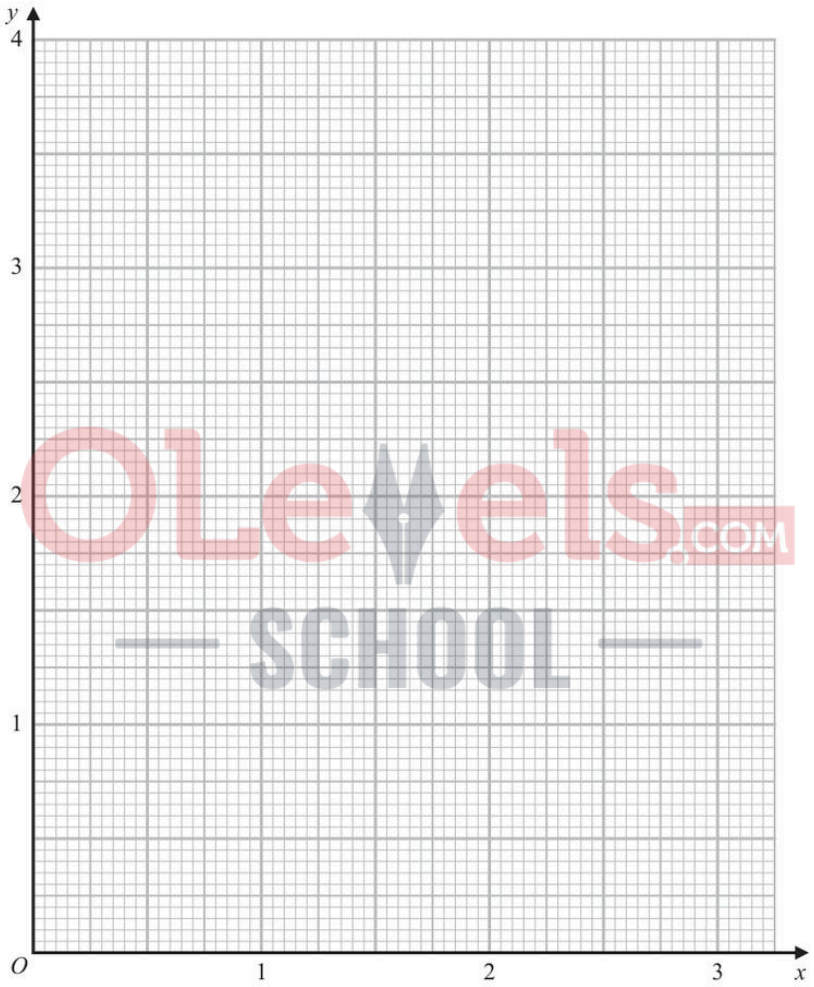
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10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

Question 6 continued



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10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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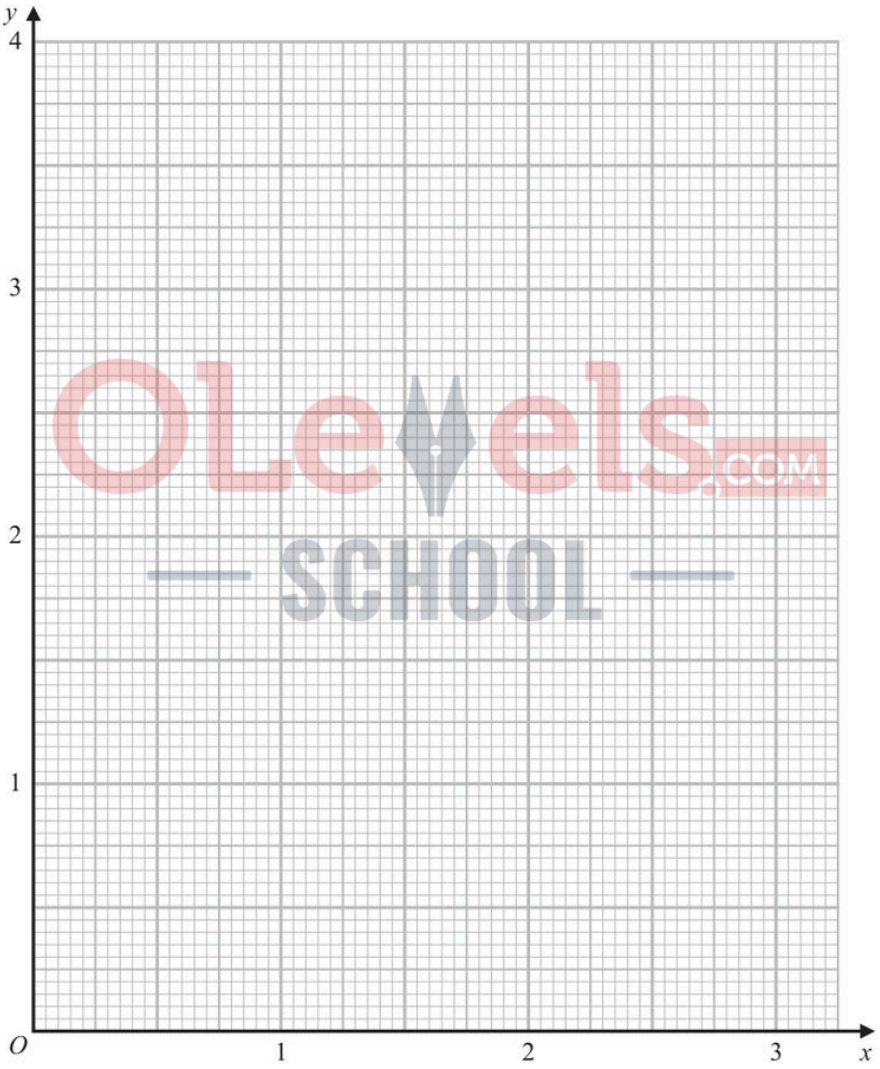
10. Question 6 · part d · 4 marks

2022 JUN · Paper 01

Further practice · Attempt part d only.

Question 6 continued

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(Total for Question 6 is 11 marks)

10. Question 6 · part d · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

4 (a) Complete the table of values for $y = \log_{10}(6x-1) - x$ giving your answers to 2 decimal places.

x	0.25	0.5	1	1.5	2	2.5	3
y		–0.20	–0.30	–0.60			

(2)

(b) On the grid opposite, draw the graph of $y = \log_{10}(6x-1) - x$ for $0.25 \leqslant x \leqslant 3$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$10^{\frac{3x-4}{2}} = 6x - 1 \quad \text{in the interval } 0.25 \leqslant x \leqslant 3$$

(4)

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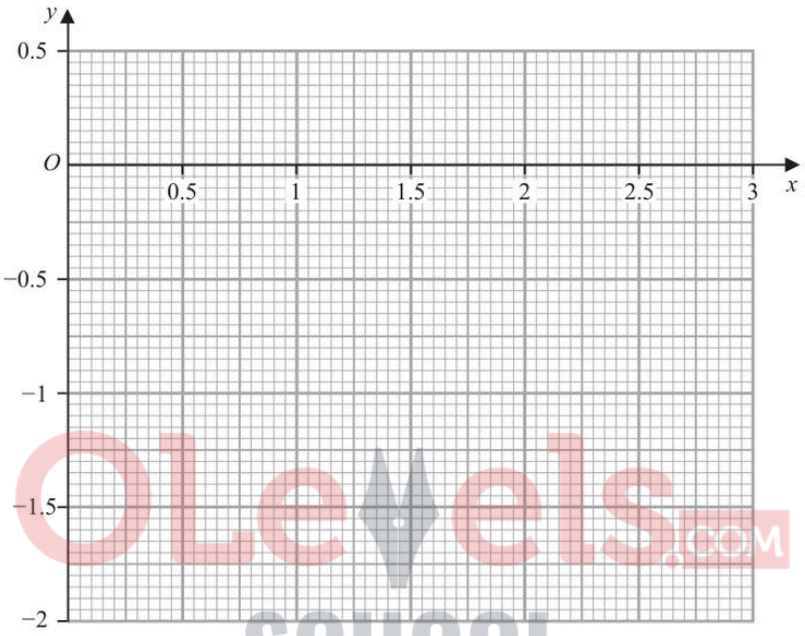
OLevels.com SCHOOL

11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued



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Turn over for a spare grid if you need to redraw your graph.

11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

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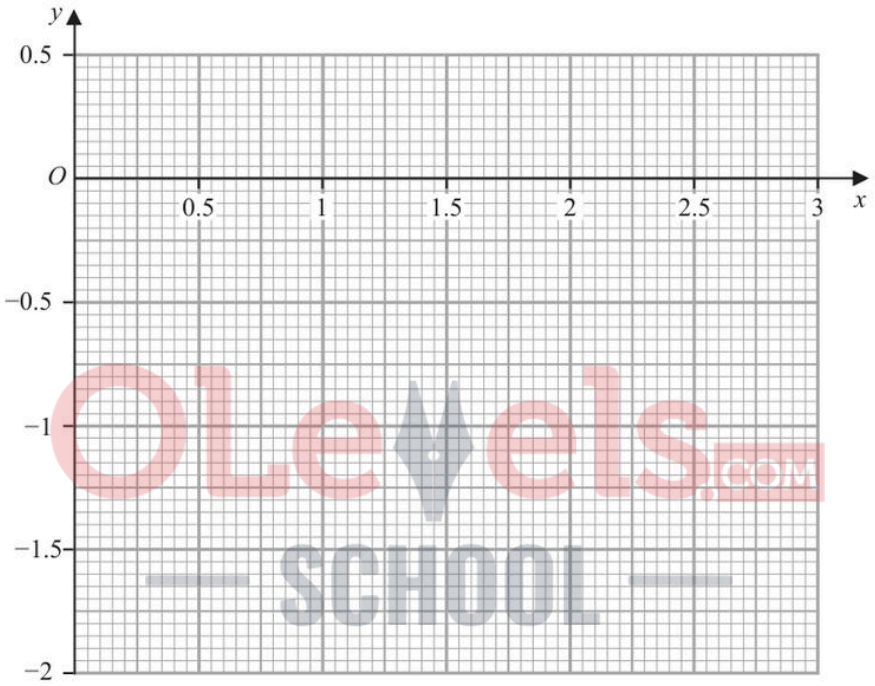
11. Question 4 · part c · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part c only.

Question 4 continued

Only use this grid if you need to redraw your graph.



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(Total for Question 4 is 8 marks)

11. Question 4 · part c · 4 marks · Working and answer

Working and answer

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12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

7 (a) Complete the table of values for $y = 3\log_3\left(x^2 - x\right) - 2x$ giving your answers to 2 decimal places.

x	1.3	1.5	2	2.5	3	3.5
y		−3.79	−2.11			

(2)

(b) On the grid opposite, draw the graph of $y = 3\log_3\left(x^2 - x\right) - 2x$ for $1.3 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$3^{\frac{4}{3}-x} = 9\left(x^2 - x\right)^3 \quad \text{in the interval } 1.3 \leqslant x \leqslant 3.5$$

(6)

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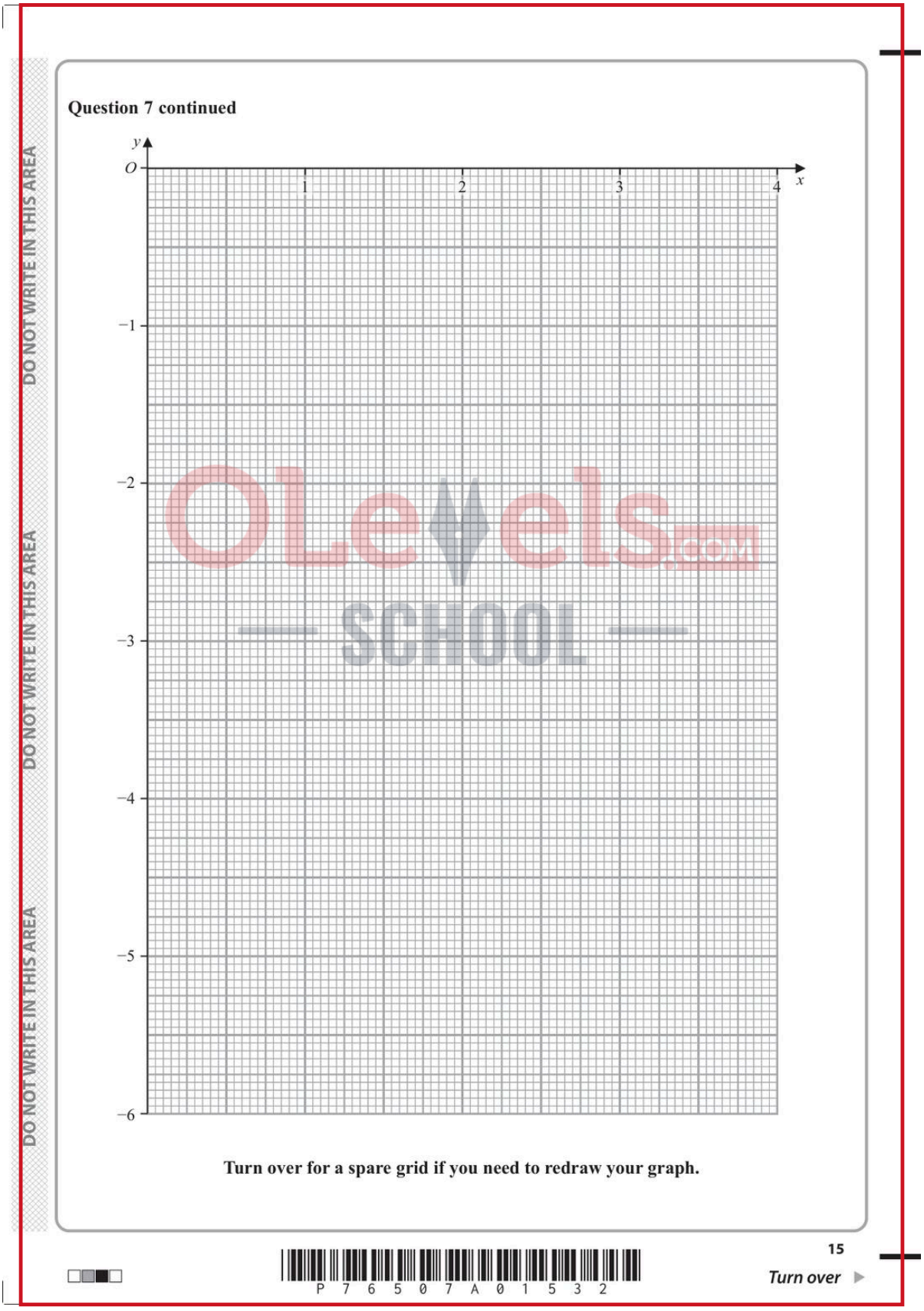
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P 7 6 5 0 7 A 0 1 4 3 2

12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.



12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

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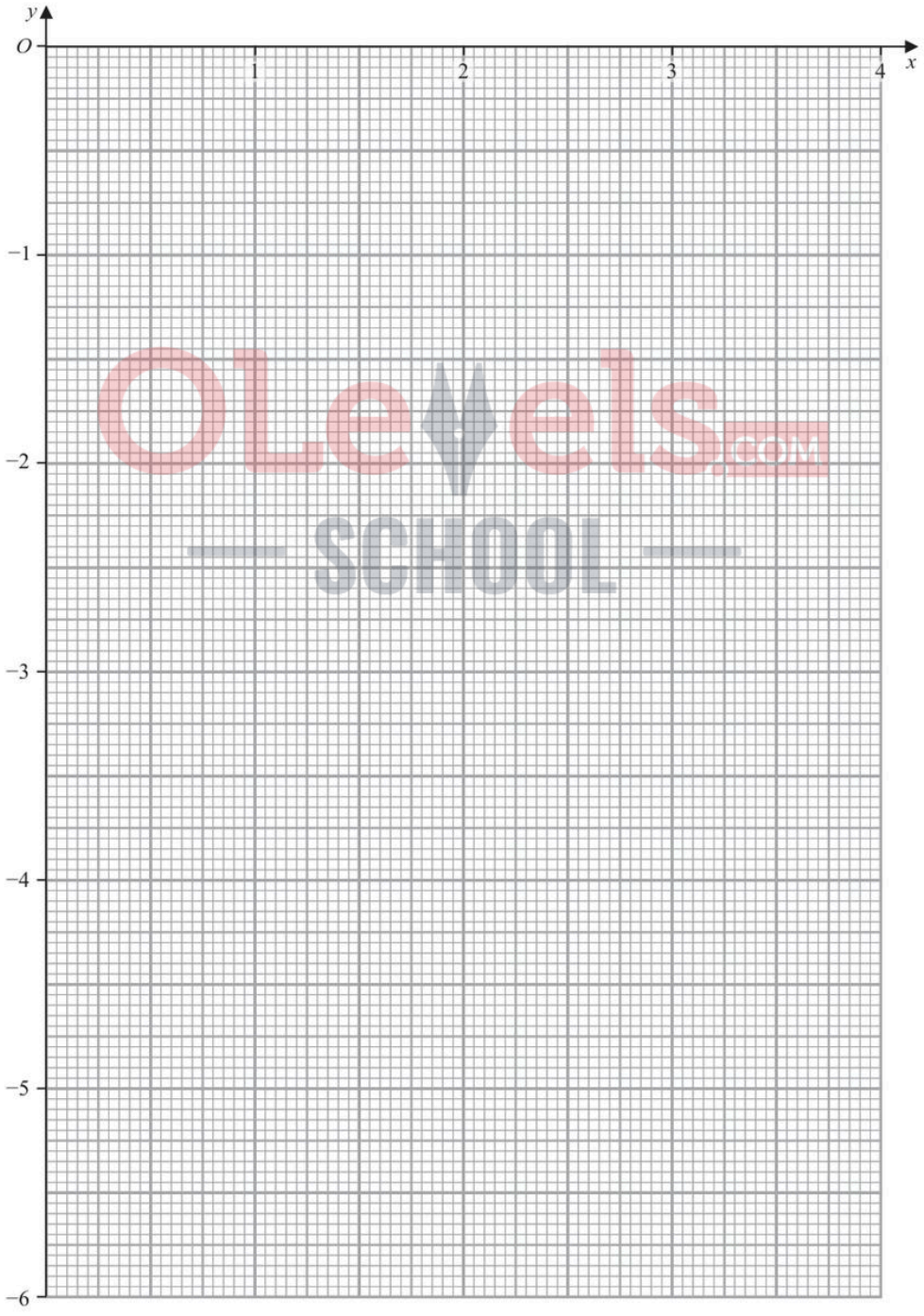
12. Question 7 · part c · 6 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 7 continued

Only use this grid if you need to redraw your graph.



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(Total for Question 7 is 10 marks)

12. Question 7 · part c · 6 marks · Working and answer

Working and answer

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